
Project Report: AI CV-Reviewer.

1. Cover Page.

- **Group Name:** Elite Group AI
- **Members' Adm no.s :** 4721, Josiah Mwangi, 5198 ,1337 and 5114
- **GitHub Link:**
<https://github.com/Pmachariah-012/wca-ai-tool-elitegroup>
- **Tool Name:** CV-Reviewer
- **Date:** 28/04/2026

2. Problem Statement.

Job seekers often face high rejection rates from automated screening systems because their CVs are not professionally optimized. While career coaching is effective, it is often too expensive for **university students, recent graduates, and unemployed youth** who are just starting their professional journeys. Furthermore, **mid-career professionals** looking to pivot and **individuals from underserved communities** often lack access to the tools needed to compete fairly.

Our tool, **CV-Reviewer AI**, provides immediate, expert-level feedback to help these candidates in the Kenyan job market improve their professional presentation. By removing the financial barrier to quality career advice, we empower **diverse talent pools** to increase their chances of success and secure the opportunities they deserve.

3. Tool Description

CV-Reviewer AI is a Python-based virtual recruitment assistant. The user interacts with the program by providing their professional experience as a

string within the script. The tool transmits this data via the **OpenAI API** to the **gpt-4** model, which analyzes the text and returns a structured critique highlighting strengths and areas for improvement.

4. AI Instruction Design (R-T-C-C-O)

The intelligence of the tool is guided by a prompt designed using the R-T-C-C-O framework:

- **Role:** Professional CV Reviewer.
 - **Task:** Review this CV for a software engineer role.
 - **Context:** The candidate has 3 years experience.
 - **Constraint:** Bullet points with feedback
 - **Output:** Highlight strengths, weaknesses, and suggest improvements .
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5. Technical Overview

The tool is built using **Python** and the **requests library** to manage API integration.

- **API Configuration:** It connects to <https://api.openai.com/v1/chat/completions> using a Bearer Token for authorization.
- **Logic:** The `connect_to_ai` function handles the POST request and uses `response.json()` to parse the message content.

- **Error Handling:** The code includes a `try-except` block to catch `requests.exceptions.RequestException`, ensuring the program doesn't crash during network issues.
- **Variables:** We used clear variable naming such as `cv_text`, `ai_response`, and `prompt` for readability.

6. Challenges & Solutions

Challenge 1 : Handling potential HTTP errors or API downtime.

Solution: We implemented `response.raise_for_status()` and a specific exception handler to return a user-friendly error message if the API connection fails.

Challenge 2 : Version control.

Solution: We used GitHub to manage contributions, ensuring multiple commits from the team to build the final `main.py` script.

7. Ethics Reflection

Privacy: The script processes data in real-time and does not store user CV text in a database or log file after the session ends.

Bias: We acknowledge that AI feedback is based on general software engineering standards and may not account for every specific local industry variation

8. Conclusion & Future Improvements

CV-Reviewer successfully automates the initial career coaching process, meeting all technical and collaborative requirements of the project. Future iterations would include the ability to upload PDF files directly and a matching feature to compare a CV against a specific job description.

9. Appendix

Full Source Code:

SUBMISSION CHECKLIST .

- ☐ **Python script (.py file) runs without errors**
- ☐ **API key is working and requests are being made**
- ☐ **R-T-C-C-O prompt is documented in the report**
- ☐ **Written report covers all 9 required sections**
- ☐ **Code has comments throughout GitHub repo created with correct naming format.**
- ☐ **Every group member has at least 3 commits on GitHub**
- ☐ **README.md is complete on the GitHub repo**
- ☐ **GitHub link is included on the report cover page**
- ☐ **Group has rehearsed the live demo at least once**
- ☐ **All group members know and can explain the code**
- ☐ **Each student has submitted individually on Google Classroom**
- ☐ **Personal reflection paragraph is included in your submission**
- ☐ **Report includes full source code in the appendix**